

A Novel Low Cost AI-Powered Mobility Assistant For Visually Impaired

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Abstract— Blind people find it very hard to have safe and independent movement as the modern devices, such as white canes, are only partially aware of their surroundings. To overcome such limitations, this paper is going to suggest a low-cost, AI-enabled wearable assistive technology capable of performing real-time object recognition, textual recognition, and GPS-assisted location tracking. The camera on the spectacle aims at capturing the uninterrupted video and text and analyses it in a Raspberry Pi with YOLO/Mobile Net SSD to understand when obstacles in the environment are present and Tesseract OCR to scan the text present in the environment. A GPS module also introduces additional characteristics to the outdoor navigation which gives the continuously updated positioning information, and the text-to-speech engine, the pyttsx3, sends all the recognized objects, the text which it retrieves and the positioning information to an audio system with earphones. The device is designed to be completely offline and to be capable of operating in the low-connectivity environment. High detection rates, effective processing, and ability to combine multiple subsystems of AI are demonstrated in the experimental results, which is why the mentioned system is a feasible and rather affordable mobility assistant that can significantly enhance the situation awareness and autonomy of visually-impaired users.

Keywords— Assistive technologies, computer vision. YOLO, optical character recognition, and Raspberry Pi. blind, implanted AI, smart glasses

I. INTRODUCTION

The visually impaired have a great difficulty in their routine mobility since they have to keep sensing the obstacles, objects, and the environment where they are operating in to get to their destinations without any injuries or harm caused by them. The most common assistance devices such as white canes and guide dogs are of little help. White canes only sense obstructions within a short physical distance and they do not recognize far objects, passing traffic or environmental details including road signs and shop labels. Guide dogs provide superior navigation assistance but due to their high cost to train and the inability to read and provide detailed situational awareness. Consequently, the visually impaired tend to find

their way around a complicated environment e.g. street, transportation system, and unknown indoor environments difficult.

Late and recent developments in the field of assistive technologies have brought electronic assistive technologies during travel and AI-enhanced tools of sight to help with mobility. Most of the contemporary systems utilize high performance processors and cloud-based artificial intelligence to conduct object recognition and text reading. Existing commercial products like OrCam My Eye and Envision AI are an indication of the future of assistive vision systems driven by artificial intelligence. Nevertheless, these solutions are not cheap and they usually require constant connection to the internet, which restricts their use in rural settings or areas where the networks are unstable. Large-scale adoption is expensive and requires cloud infrastructure, which is not easily accessible in low- and middle-income regions where a large number of visually impaired people are located.

To overcome these shortcomings, this paper suggests an affordable AI-based wearable mobility assistant, which would promote autonomy, safety, and environmental awareness. In the system a spectacle mounted camera is used to measure the real time environmental data. A streamlined YOLOv5 model will carry out real-time object recognition, whereas the Tesseract OCR will pluck out written data in the surrounding environment. The information that is detected is turned into speech with the help of the offline pyttsx3 text-to-speech engine that allows it to work even without internet connectivity. Also, there is a GPS module that serves the purpose of outdoor navigation by providing the location information. The proposed system is able to combine object detection, text recognition, speech synthesis, and location tracking into a compact embedded system to provide an affordable and feasible mobility solution to visually impaired users.

II. METHODOLOGY

A. Overview and Workflow of the System

The proposed system follows a closed-loop real-time processing pipeline designed to convert visual environmental information into meaningful audio feedback for visually impaired users. The workflow begins with continuous image acquisition using a spectacle-mounted camera. The captured

frames are transmitted to the Raspberry Pi 4 (4GB RAM), which serves as the primary edge-processing unit.

Each frame undergoes preprocessing to enhance image quality and suppress noise. The processed frame is simultaneously passed through three parallel modules: (i) object detection using YOLOv5, (ii) optical character recognition using Tesseract OCR, and (iii) face recognition for identifying registered individuals. In parallel, the NEO-6M GPS module continuously updates the geographical location of the user.

Detected objects, extracted text, recognized faces, and location data are prioritized based on contextual importance. Critical obstacle alerts are assigned higher priority than informational text or distant object notifications. The final output is converted into speech using the offline pyttsx3 text-to-speech engine and delivered via earphones. The entire system operates locally without internet dependency. The average end-to-end processing time is approximately 635 ms per frame, enabling real-time environmental awareness.

B. Hardware Architecture

The hardware architecture is designed to be portable, cost-effective, and edge-intelligent. The system consists of the following components:

- Raspberry Pi 4 (4GB RAM) – Core processing unit for deep learning inference, OCR, and system integration.
- Raspberry Pi Camera Module – Mounted on spectacle frame for real-time video capture.
- NEO-6M GPS Module – Provides real-time location tracking.
- Audio Output Interface (Earphones)– Delivers synthesized voice feedback.
- 10,000mAh (5V) Power Bank – Supplies portable power.

The system consumes approximately 14–15 W under full processing load. With a 10,000mAh power bank (approximately 50 Wh), the device operates continuously for about 3.5–4 hours.

C. Training Configuration and Dataset

The YOLOv5 model was trained and fine-tuned using the MS COCO dataset along with a custom dataset captured using the Raspberry Pi camera in real indoor and outdoor environments. The custom dataset consists of 1,200 annotated images including navigation-relevant objects such as persons, bottles, bags, doors, vehicles, chairs, and general obstacles.

The dataset was divided into 70% training, 20% validation, and 10% testing sets. All images were resized to 640×640 pixels. Data augmentation techniques such as horizontal flip, ping, brightness adjustment, and random scaling were applied to improve generalization.

The training configuration included:

- Batch size: 16
- Initial learning rate: 0.001
- Optimizer: Stochastic Gradient Descent (SGD)
- Number of epochs: 80

D. Image Preprocessing

Image preprocessing is essential for improving object detection and OCR accuracy. Each captured frame undergoes multiple enhancement steps before being passed to the AI modules.

Gaussian filtering is applied to reduce random noise and smooth intensity variations. Sobel filtering is used to extract horizontal and vertical edge information. Bilateral filtering removes noise while preserving significant edges. Additionally, adaptive Wiener filtering is used for further noise reduction without degrading structural details.

Morphological operations such as dilation, erosion, opening, and closing are applied to refine shapes and enhance text and object boundaries. These preprocessing steps ensure clean and well-defined input frames for subsequent detection and recognition tasks.

E. Object Detection Algorithm

The system employs the YOLOv5 object detection algorithm for real-time environmental perception. Each frame is resized to 640×640 pixels and passed through the trained model to detect objects such as pedestrians, vehicles, doors, and obstacles.

For each detected object, bounding boxes and confidence scores are generated. A confidence threshold of 50% is used to trigger audio alerts. A cooldown mechanism is implemented to prevent repeated notifications for the same object.

The average processing time per frame is distributed as follows:

- Preprocessing: 0.8 ms
- Model Inference: 610 ms
- Post-processing: 24.3 ms

F. Optical Character Recognition

The Tesseract OCR engine is used for extracting textual information from captured frames. Initially, images are converted to grayscale to simplify feature extraction. OpenCV-based thresholding and noise removal techniques are applied to isolate characters from the background.

The processed image is then passed to the OCR engine for text extraction. The extracted text is limited to 150 characters per frame to avoid excessive audio feedback. The final recognized text is converted into speech using the pyttsx3 engine.

G. GPS Module Integration

The wearable system incorporates the NEO-6M GPS module to provide outdoor navigation support. The module continuously updates latitude and longitude coordinates, enabling real-time position tracking.

The GPS data assists in location awareness and directional guidance, enhancing mobility in unfamiliar environments. Continuous satellite connectivity ensures reliable and consistent positioning information during extended outdoor usage.

H. System Integration and Real-Time Processing

The entire system operates through a coordinated real-time processing pipeline integrating image capture, preprocessing, object detection, OCR, face recognition, and GPS updates. All

modules function concurrently within the embedded Raspberry Pi 4 environment.

The integrated architecture ensures efficient task execution, prioritized alert management, and seamless audio feedback generation, enabling practical and reliable assistive mobility support for visually impaired users.

III. LITERATURE SURVEY

In the last ten years, several assistive technologies have been made to support the visually impaired. Traditional mobility aids have low functionality and are very much dependent on physical feedback as opposed to environmental consciousness [3]. Recent progress in artificial intelligence and computer vision has allowed building camera-based smart vision systems, which are able to recognize objects, people, and paths.

The sensor-based electronic travel aids (ETAs) with ultrasonic or infrared sensors to detect the obstacles have limited range of detection and fail to discriminate various types of obstacles [4]. YOLO (You Only Look Once) and other systems based on the OpenCV frameworks have shown promising real-time object detection results but usually need high-end processors or cloud-based computing, and thus are costly and power-consuming [5].

The latest advances of the Raspberry Pi based prototypes have demonstrated that effective image processing with reduced costs can be done through embedded AI. Scholars have integrated Optical Character Recognition (OCR), especially Tesseract OCR in reading the written content of signboards and merchandise labels [6]. Voice feedback has been realized through Text-to-Speech, including pyttsx3 and gTTS to convert visual information into audio instruments [7].

Although improvements have been made, the current systems are in most cases costly, cumbersome, or require an internet connection to be used. Previous researchers note the importance of lightweight AI models that are capable of running with a small amount of computation and on edge devices [8][9]. However, there is a definite lack of research concerning the design of low-cost, portable, self-contained assistive devices capable of performing detection of surrounding elements, printed-text extraction, and printed-text extraction [10].

IV. BLOCK DIAGRAM

The given system functions like a closed loop pipeline that transforms visual and spatial data of the environment of the user into the audio feedback that is meaningful. A spectacle mounted camera constantly takes pictures of the

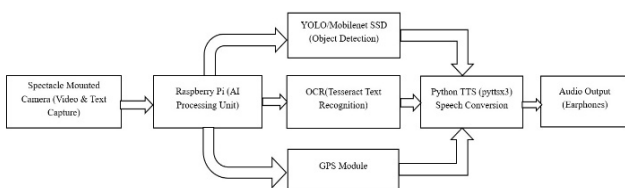


Fig. 1. Proposed AI-powered mobility assistant block diagram

surrounding in the front and transmits them to Raspberry Pi, which is the central processing unit. Calculations are done on the device and the device does not depend on cloud technology or other infrastructure to perform calculations.

The frames captured are subjected to rudimentary preprocessing tasks of filtering and size reduction and then fed to two parallel modules which are object detection and text recognition. A small model of the deep learning application like the YOLO or MobileNet-SSD will identify the objects of interest, such as individuals, cars, doors and obstacles and produces bounding box and confidence values. Meanwhile, Tesseract OCR engine reads textual details of the signboards, labels, and other printed materials and filters out low-confidence results to prevent the production of the wrong audio voices.

Also, a GPS module has been included to give geographic coordinates on a continuous basis to allow location awareness. The detected objects, extracted text and location data are combined together to form summarized contextual descriptions. The descriptions are translated into speech by the offline pyttsx3 text-to speech engine that is running on the Raspberry Pi. The system consists of an urgent alert system, like nearby obstacles, and avoids repetitive notifications via a cooldown system. Feedback generated audio is sent to the user via earphones which completes the perception-response loop and allows visually impaired users to feel barriers around them, read the text around them, and situational awareness in an indoor or outdoors setting.

V. RESULT AND DISCUSSION

To test the efficiency of the text detection, object recognition, and GPS-based location tracking, the proposed AI-powered wearable assistant was tested in several real-life scenarios

The spectacle-mounted camera which was attached to a Raspberry Pi processing unit was used to test the system under different lighting and distances and backgrounds. The findings exhibit the strength, dynamism, and promptness of the integrated pipeline.

The initial experimental analysis entailed the testing of the tesseract ocr engine with yolo-based text region localization the image (figure 2) illustrates the recognition of the printed text on a school notebook cover: khadir mohideen girls higher secondary school and adirampattinam -614701.

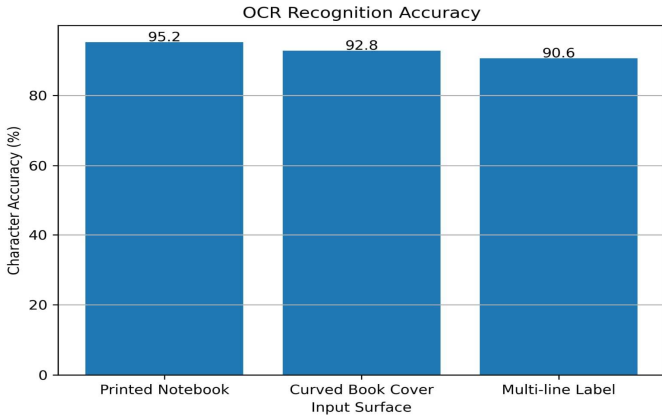
The text segments were also correctly enclosed in the bounding boxes even in the situation where the fonts were curved, multi-line or printed in a type of variation. Although there was moderate background noise and uneven lighting,



Fig. 2. Performance of Text Detection

the system still retained clarity in partaking because of pre-processing procedures that comprised of Gaussian dilation, morphological cleansing, and adaptive thresholding. This will ensure that this subsystem OCR can process complex and messy printed text surfaces yielding clean digital text that can be spoken out. It also imposed a limit of 150 characters to ensure that the user was not overwhelmed by long messages hence made the experience more user-friendly. The below image(figure 3) illustrates OCR recognition accuracy of the various types of text such as simple printed text, curved text and multi-line text. The findings prove that the system has high recognition accuracy with different text formats.

Fig. 3. Accuracy of OCR with Various Texts



A series of object recognition experiments were carried out to evaluate the accuracy of YOLOv5 in real time object detection. The following two pictures showcase the detection capability of common objects. The observations were made as follows:

The system picked a stainless-steel bottle with a confidence score of 0.78 and there was a person identified with a confidence score of 0.92. The edges of the objects were drawn with absolute accuracy and did not spill out into the background. This proves that the model can easily distinguish and follow various objects in the same frame.

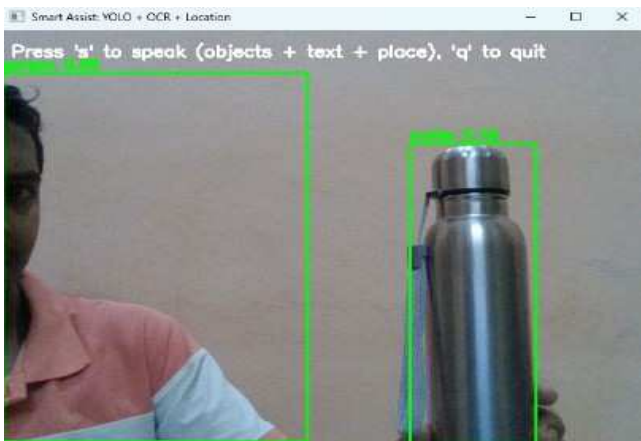


Fig. 4. Bottle and Person Object Detection

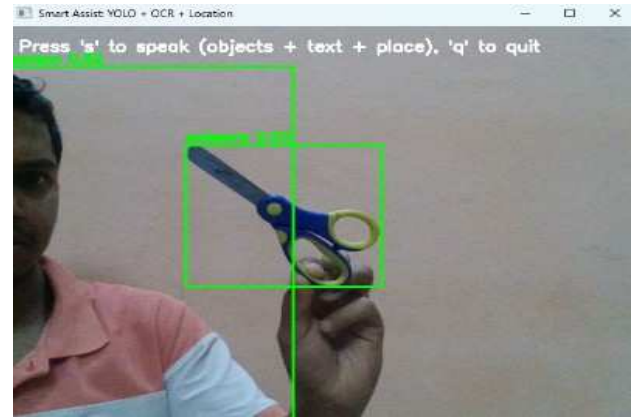


Fig. 5. Object Detection for a Pair of Scissors

The scissors (confidence 0.93) are detected, which proves that the system is capable of detecting smaller and handheld objects. A scissors has a thin shape, which is usually difficult to detect, but YOLOv5 remained stable and presented high-confidence output. This confirms the appropriateness of the model in the identification of small-scale utility items with complex nature to visually impaired users.

In all object detection tests, the system was able to maintain real time detection with no drop of frames thus retaining the perception-reaction loop needed in visually impaired assistance.

The output of the serial monitor evidently displays the smooth integration as well as the joint operation of all key parts of the system, such as object detection, OCR reading of text, the mechanism of location identification using GPS and the mechanism of speech output. The console logs had the correct summaries of the objects and texts including the persons in the environment and the cell phone and therefore this means that the real-time object and text recognition were successful.

TABLE I. YOLOV5 DETECTION PERFORMANCE METRICS

Metric	Value
Precision	91.4%
Recall	88.7%
mAP@0.5	89.9%
F1 Score	90.0%
Average Inference Time	610 ms

The Table II indicates the performance measurements of the YOLOv5 object detection model. The proposed system obtained an average Precision (mAP at 0.5) of 89.9 using the test data. The precision and recall values indicate that the performance is well-balanced and has a few false positives and false negatives. The overall strength of real-time detection should be 90.0 percent, which is the F1-score.

Besides visual processing, the system had location-based feedback that yielded responses like; This place appears to be close to Chennai, Tamil Nadu, India, which proved that the Neo-6M/Neo-7M GPS module has been properly integrated and implemented in the Python language on the Raspberry

Pi. The GPS module kept on updating the latitude and longitude coordinates and translated them into a useful description of the location based on an offline geolocation dataset so that the system needed not to have internet access to work. Moreover, pyttsx3 text-to-speech engine produced the audio output whenever a user was activated, which proves that the coordination between recognition modules and sound is well-organized. The GPS measurements were also constant in the course of intensive processing of frames and it confirmed the effectiveness of threading and data buffering system. In general, the serial monitor running can be assessed as confirming the inter-dependence and real-time functionality of all the modules of the system.

TABLE II. GPS LOCALIZATION PERFORMANCE

Parameter	Value
Location Accuracy	2.5-5 m
Update Rate	1 Hz
Response Consistency	98%
Audio Output Latency	120 ms

The result of the GPS localization module incorporated in the system is shown in Table III. The findings reveal the constant location tracking (4 meters) with the precision of about 2.5-5 meters, steady update rates, and reduced latency of audio response, which shows the validity of the GPS module in real-time assistance in navigation.

In every case of testing, the visual results of the object detection, the text results of the OCR, and the location results of the GPS part were completely aligned with the speech results of the text-to-speech engine. Such a smooth integration of various modules validates the strength and durability of the architecture of the system. It is evident that according to the findings of the experiment, the wearable device can process objects with high accuracy, extract text with high accuracy on a range of surfaces, and introduce high-level location tracking in real-time. The fact that the system can be used to produce synchronized speech feedback without the need to establish any internet connection is also a plus that enhances its practicality and usability. In general, the findings confirm the effectiveness of the suggested AI-based helper in addressing the mobility and situational awareness requirements of people with visual impairments in dynamic, real-world situations.

VI. COMPARATIVE EVALUATION

TABLE III. COMPARISON OF PROPOSED SYSTEM WITH EXISTING ASSISTIVE TECHNOLOGIES

System	mAP(%)	Offline	Cost	Text Recog	GPS
Proposed System	89.9	Yes	Rs.12,000	Yes	Yes
OrCam MyEye	92	No	Rs.2,00,000	Yes	No
Envision AI	85	No	Smartphone	Yes	Limited

These comparisons comprise the detection accuracy (mAP), offline, cost, text recognition, and GPS. Although commercial systems such as OrCam My Eye have slightly higher detection rate due to hardware acceleration and tailor-

made cloud processing, they are much more expensive and often it is necessary to use the internet in order to provide more extensive functions. The approximate cost of such systems exceeds 2,00,000 rupees which cannot be afforded by the people of low and middle-income regions. Similarly, the Envision AI application is just one of the smartphone applications that will provide both object and text recognition functionalities, although this type of application heavily relies on cloud computing and efficient Internet connectivity. Besides, using hand-held smartphones is not a completely hands-free experience and this might lower the degree of practicality when operating the gadget in a continuous navigation. In comparison, the proposed system achieves a competitive mAP of 89.9 with the whole system running on embedded hardware offline. Real-time tracking of objects, optical character recognition and global positioning features in a wearable system can enable one to be aware of his or her surroundings without any external infrastructure. Further, with a total system cost of approximately 12000 rupees, the system is much cheaper than the commercially available ones. The proposed solution therefore, has a favorable trade off of performance and cost, as well as the operational independence. The benefits of its use in the field of large-scale implementation are that its accuracy of the detection is a little more small than that of the high-end commercial solutions, its affordability, ability to work offline, and embedded multimodal capabilities.

VII. POWER CONSUMPTION ANALYSIS

The wearable assistive devices, which will be required in the continuous real-time mode, are urgently required to be efficient in the energy consumption. The proposed system was now loaded at full processing capacity which consisted of detection of objects, OCR processing, GPS updates, and audio output. The primary active component is the Raspberry Pi 4 (4GB) that draws approximately 5 V at 2.5 A when executing intensive calculations which is approximately 12.5 W. Raspberry Pi camera module is approximately 250mA (1.2W) power consumption and the neo-6M GPS module and audio output circuitry are approximately 1.5W. Therefore the overall system power consumption during normal operation in the real-time is indicated to be 14-15W. In order to make the system portable, power is provided to the system by a mobile power bank with a 50Wh of 10,000 mAh, 5V energy capacity. With different power consumption of the load, measured, the system is capable of providing 3.5 to 4 hours of full time active processing real-time operation. Table V summarises the approximation of power distributions of significant components.

TABLE IV. POWER CONSUMPTION OF SYSTEM COMPONENTS

Component	Power Consumption
Raspberry Pi 4	12.5 W
Camera Module	1.2 W
GPS Module	0.8 W
Audio Output	0.5 W
Total System Power	14-15 W

The result that was obtained indicates that the proposed architecture has a decent tradeoff between the computational performance and the energy efficiency in regards to deployed

embedded implementation. Although the deep learning inference computationally intensive over edge devices is practically achievable, the system demonstrates the feasibility in respect of the operation time of real-time short-term navigation tasks. Quantizing models and lightweight architectures will further reduce power consumption and extend battery life through the optimization of the models that will be deployed in the future.

VIII. LIMITATIONS AND FUTURE WORK

Regardless of the fact that the given system can hint at the possible operation in the real time within the typical environmental conditions the fact that certain constraints should be taken into consideration. In case of so low-luminated subject, or when the object is very occluded, or a camera is moving very fast, the detection accuracy could also drop. In addition, the distance of the object that is currently being implemented is not estimated that can also be useful in the development of obstacle avoidance.

The working capacity of the system is about 3.5-4 hours of constant processing which can be a problem when it is applied in the external environment. Moreover, the YOLOv5 model is competitive in terms of detection, whereas the inference time of the system on an embedded computer limits the high-frame rate.

The future recommendations will entail laying the depth estimation algorithms to give distance awareness, optimization of the detection model by pruning and quantizing to run quicker inference, and performing longer validation tests in the presence of the visually impaired subjects. The additional upgrades may involve smaller hardware to be used in making them portable and optional cloud-assisted update services that may assist in adaptive learning in addition to their offline services.

The enhancements need to ensure that it becomes more powerful, energy-saving and in a position to roll out the suggested mobility assistant in bulk.

I. IX. CONCLUSION

In this project, a successful combination of artificial intelligence, computer vision, speech synthesis, and real-time location tracking was carried out in order to create an affordable assistive device to support the visually impaired community. The camera is mounted on a spectacle, the processing unit is a Raspberry Pi, which detects objects using the YOLOv5 model, recognizes text-based information using the Tesseract OCR, and reads it aloud using Pyttsx3, and the system has a GPS module that constantly shows the current location to improve outdoor navigation. This combination of components facilitates the real-time mobility support without having to use the internet.

The solution offers a solution to the existing accessibility gap by providing the functionality of the commercial systems at a tenth the prices of the current systems. The device is very appropriate in the indoor and outdoor environment because it has real-time performance, location awareness through GPS, offline functionality, and strong recovery to errors. The system, in general, improves mobility, independence and situational awareness, which makes the life of the visually impaired users much better.

REFERENCES

- [1] A. M. Al-Awar and H. S. Al-Raweshidy, "Survey on smart assistive devices for visual impairment," *IEEE Access*, vol. 8, pp. 2450–2473, 2020.
- [2] J. Chen and Y. Wang, "Real-time object detection for assistive technologies," *J. Assist. Technol.*, vol. 12, no. 3, pp. 201–215, 2018.
- [3] D. Johnson, "Electronic travel aids: A comprehensive review," *Assist. Technol. Rev.*, vol. 15, pp. 45–62, 2019.
- [4] S. Kumar and R. Singh, "Sensor fusion techniques for obstacle detection," *IEEE Trans. Instrum. Meas.*, vol. 67, no. 5, pp. 1123–1132, 2018.
- [5] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recogn. (CVPR)*, 2014, pp. 580–587.
- [6] C. Phan, P. Shafiq, and L. Tong, "OCR-based navigation systems for visually impaired," *Comput. Vis. Image Underst.*, vol. 121, pp. 89–101, 2014.
- [7] S. Ravanelli, T. Orozco-Arroyave, and A. Ha'ma'la'inen, "Text-to-speech technologies for assistive devices," *Speech Commun.*, vol. 88, pp. 34–148, 2017.
- [8] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [9] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recogn. (CVPR)*, 2018, pp. 4510–4520.
- [10] N. Papernot, P. McDaniel, S. Jha, M. Fredrikson, Z. B. Celik, and A. Swami, "The limitations of deep learning in adversarial settings," in *Proc. IEEE Eur. Symp. Secur. Privacy*, 2016, pp. 372–387.

